

Research

BIM-based energy consumption analysis for residential buildings: effects of orientation, size, and type of windows

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Abstract

Buildings have a significant environmental impact due to their reliance on nonrenewable energy sources. To address this concern, computer software is utilized to estimate building energy requirements. Building Information Modeling (BIM) has emerged as a valuable tool in this process, with BIM 6D offering a particularly beneficial feature. BIM 6D provides a comprehensive energy model of the building, simulating its actual energy performance. By analyzing natural and artificial lighting systems, particularly daylighting, BIM 6D can assist in optimizing energy efficiency. Despite being underutilized, the potential of this feature to enhance building sustainability is noteworthy. BIM 6D facilitates well-informed design and operational decisions for both new constructions and the renovation of existing buildings. The study examines design criteria such as building orientation, window orientation, size, and type to optimize energy consumption in a bungalow in the Sangli region of Maharashtra, underscoring the importance of energy-efficient buildings to local communities. Energy analysis and optimization are conducted using the Autodesk Insight tool. The methodology involves creating 3D models with Autodesk Revit Architecture software and performing various analyses, including Building Energy Modeling (BEM) simulations, Illuminance Analysis, Solar Access, and Daylight Autonomy assessments. The analysis findings indicate that low-e glass windows of small size and oriented between 0° and 90° significantly enhance energy efficiency under certain conditions, while changing the window type does not affect the analysis. The paper concludes with recommendations for window orientation and size to enhance energy performance in residential buildings, contributing to the broader goal of sustainable and energy-efficient construction practices. This study provides valuable insights for designing energy-efficient residential buildings in rural areas, addressing awareness gaps, and promoting sustainable construction practices.

Keywords BIM · Sustainability · Energy analysis · Residential building

1 Introduction

1.1 Background

The International Energy Agency reports that buildings consume approximately 35% of the world's total energy, posing a significant challenge to global sustainability [1]. New building-integrated ideas, such as solar passive and mixed-weighted thermal mass structures, and efficient energy technologies are being invented to deal with the increasing need for energy in buildings along with environmental issues [2]. Energy-efficient structures are gaining importance

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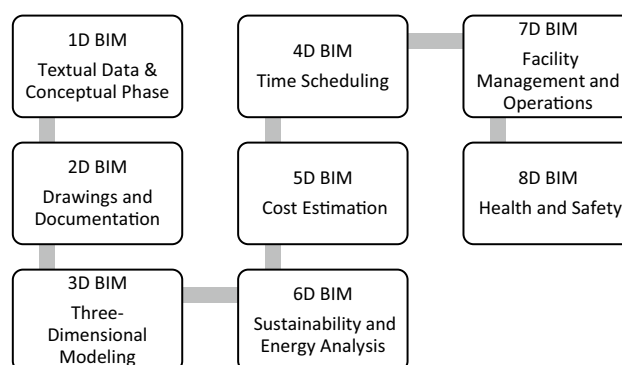
due to the continuous increase in greenhouse gases and energy consumption as energy efficiency leads to structures with low cost suitable for environmental conditions and can fasten the construction process[3]. According to today's scenario building sector contributes more to energy consumption and CO₂ emissions globally [4]. Because of this, the importance of sustainability is increasing rapidly [5]. In some countries, funding is also provided to implement various sustainable techniques and to motivate their use [6]. The use of BIM is also increasing rapidly to increase new trends and implement multiple techniques to make structures more sustainable. Through digital prototyping and simulation technologies, the invention of building information modeling (BIM) technology offers a new way to predict, handle, and analyze the environmental effects of project construction and progress [7].

1.2 Need

BIM dimensions (Fig. 1) are advantageous in every stage of the construction process. Dimension 1D and 2D include texts, sketches, pictures and floor plans, sections & elevations, etc. which help to make 3D models [8]. BIM uses 3D CAD models as a centralized & useful database for all facility information like the life-cycle approach, and operational, and technical information. This combined environment leads to computerized work procedures and virtual simulations that boost productivity and decrease costs [9]. The time was introduced as the 4D and cost as the 5D in BIM, which extended beyond 3D [10]. The use of a 4D model gives a better visualization & scheduling to see whether the construction activities going on site are as per plan or not which would be helpful for the engineer as well as the contractor [11]. The 5D facilitates actual cost budgeting and performance analysis of active projects, which generates cash flow at various project phases [12]. The energy conservation and environmental investigations & recreations conducted during the project stage and occupation are included by the 6D while the management and maintenance of the building is covered by the 7D and the human safety in construction is covered by the 8D BIM model, which supports preventive care actions like identifying risks in construction [13]. There are various softwares available to fulfill the requirement of these BIM dimensions which helps to make the structure more sustainable. Software like Energy Plus, eQuest (DOE-2.2 engine), DOE-2, Design Builder (EnergyPlus engine), Revit Architecture with Insight plugged in, Green Building Studio (DOE-2 engine), etc. used to make energy simulation easy and can align the model with energy standards and codes [14]. As we can see, the energy consumption of the 3D model used in this research case study has been calculated (Fig. 2).

When BIM and BEM are used together, architects can create more effective solutions that help owners keep their buildings operating at maximum efficiency for a longer time, and builders can create more effective building processes that open the way for a more sustainable future [15]. Energy assessment tools such as Autodesk Insight may be used to study the different possibilities [16]. The necessary changes in the building model & energy consumption can be done in the software to find the best option for energy efficiency [17]. Daylighting autonomy and Illuminance analyses can be done with Plugged Insight in Autodesk Revit Architecture, as the presence of daylight and direct sunshine are crucial components of the architectural experience [18]. It is possible to reduce the building's energy consumption by improving factors like window size and type, building orientation, etc. [19]. Despite the advancements in BIM and the government's efforts to promote energy-efficient construction, many Indian villages still lack knowledge about these methods, prompting us to conduct a survey with local engineers to assess current conditions and practices in energy-efficient bungalow construction.

Fig. 1 BIM dimensions



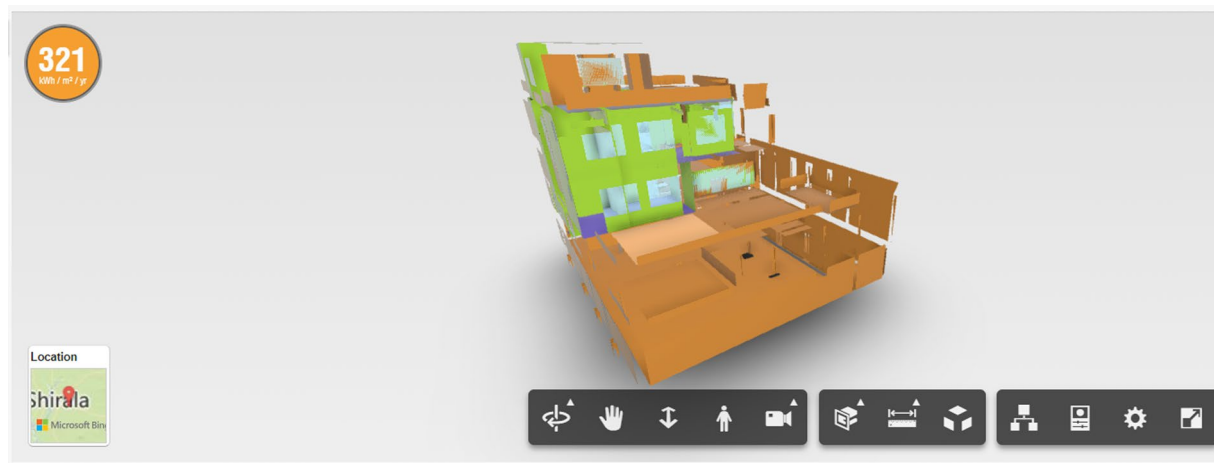


Fig. 2 Building view and outcome displayed on the INSIGHT 360 platform

1.3 Aim of the study

This paper aims to demonstrate the importance of energy-efficient buildings by analyzing a constructed bungalow to highlight sustainability and energy savings. It investigates the impact of window orientation, size, and type on energy consumption in residential buildings using Building Information Modeling (BIM), analyzes how these factors influence overall energy efficiency and identify optimal window configurations to enhance energy performance. The study provides insights and recommendations for designing energy-efficient residential buildings.

2 Literature review

2.1 Climate change and energy use

The primary hazards to civilization, associated with climate change and global warming, are directly connected with energy use and greenhouse gas emissions [4]. The International Energy Agency has unequivocally reported a troubling trend in global energy usage and carbon emissions over the past two decades (1984–2004). Primary energy consumption surged by 49%, while carbon dioxide emissions skyrocketed by 43%. This translates to an average annual increase of 2% for energy and 1.8% for carbon emissions [20]. Approximately 25% of CO₂ emissions and 30% of world energy use are related to buildings [21]. Many international institutions encourage energy efficiency as the best tool to manage energy demand and transition to a low-carbon future, leading governments to emphasize and promote it through various policies [22].

2.2 Studies on energy efficiency and building design

Many studies have aimed to reduce energy consumption and develop and implement various strategies to find optimal solutions. For instance, a case study of a multi-story, multi-family residential home in Afghanistan provided a simplified energy analysis to evaluate the impact of the building's orientation. This study identified the optimal and worst orientations and assessed the associated cost savings [23]. Using a 3D model of an actual building located in Hertfordshire, CM23 5NW, UK, F.H., authors discovered that a well-oriented structure may save significant quantities of energy during its entire lifespan and replicate its energy efficiency [24]. Daylight plays a crucial role in shaping the built environment, influencing factors such as space size, visual comfort, and thermal performance. By reducing the reliance on artificial lighting and regulating internal temperatures, daylight contributes to creating more sustainable and energy-efficient interiors. On the other hand, daylight as a significant design variable has a fundamental impact on building form, orientation, geometry, and quality of interior space (Ander 2003). At a specific latitude, building occupants can take advantage of sufficient natural daylight entering through windows, thereby preserving both the benefits of natural light and visual content. A method for designing energy-efficient buildings was proposed,

emphasizing the initial design stage due to its significant impact on environmental conditions and energy performance. This approach involves analyzing the content of each component technology and the existing construction process [25]. The BIM methodology enables us to comprehensively analyze the relationship between lighting levels, occupant comfort, and energy efficiency. By simulating various design factors, such as building orientation, window size and type, and lighting equipment consumption, we can optimize the design to achieve optimal daylighting and minimize energy usage. This holistic approach allows for informed decision-making and the development of more sustainable and energy-efficient buildings.

2.3 Software for energy simulation

Industry developing much software in BIM which is very helpful in Energy simulation to find the energy consumption of the building. Figure 3 displays a variety of simulation software. According to research, Virtual Environment™ appears to be the best and most flexible solution for environmental sustainability evaluations. While it is superior to Green Building Studio™ in multiple areas, Ecotect™ is considered less adaptable of the three [26]. One of the studies indicated that comprehensive information on optimizing energy use was supplied by the project research in the Insight 360 program [27]. Using Autodesk Revit and Autodesk Insight software, it was concluded that a BIM model accurately represents the need to address non-renewable energy consumption by shifting to solar radiation systems. Additionally, proper building orientation can reduce electricity bills, which is particularly beneficial for Indian families in medium- and low-income groups [28]. With Revit Insight plugged in and the sample shown in Fig. 4, one may do several kinds of sun path-based analysis.

An energy consumption analysis using Autodesk Revit 2017 and Green Building Studio software found that adding ceiling tiles to a building significantly enhances energy savings, reducing consumption by nearly 20%. Wall type follows, contributing to over 10% in energy savings, while windows with single-glazed glass affect 7% of energy usage, making them the third-highest factor [29].

2.4 Factors influencing energy consumption

Several case studies and research have been done which include key factors influencing energy consumption in residential buildings including building design, materials, insulation, HVAC systems, and occupant behavior, with a specific focus on windows as a critical factor affecting thermal performance and energy use. Three major categories of internal and external impacting elements were recognized by Shuo Chen et al. These categories included occupant behaviors, building features, and equipment and technology [30]. It is believed that several family characteristic components impact the behavior of residential electricity users [31]. Building characteristics, such as orientation, floor plan depth,

Fig. 3 Revit insight model view of analysis

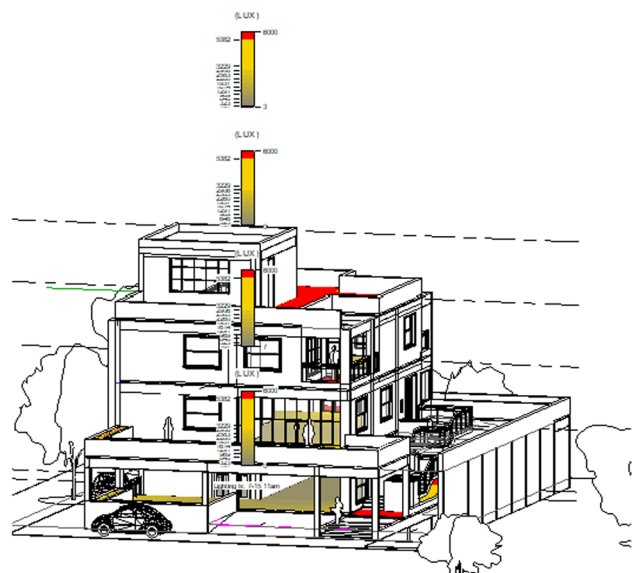
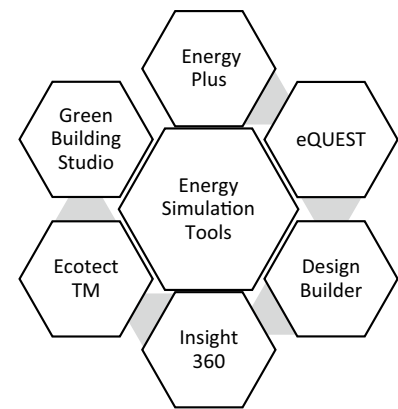


Fig. 4 Energy simulation softwares



space utilization, window area, and glazing type, have a significant impact on a building's energy performance during the early design phases. A study showed that the way windows are opened significantly affects the energy used to maintain indoor environmental quality and the quality of the interior environment [32].

2.5 The impact of window design on energy consumption

With the various study-related window orientations, positions, etc. its glass type also affects the major on consumption of energy. Window glazing significantly impacts the total energy use and efficiency of a structure. The transfer of heat through glass windows from incoming sunlight assists in heating, but it also increases the demand for air conditioning [33]. To enhance energy efficiency, reduce greenhouse gas emissions, and lower power costs in a commercial building located in Pune, specific design elements were evaluated. The study found that a building orientation of 90 degrees, a window-to-wall ratio (WWR) of 15%, and the use of triple Low-E glass are ideal for this region. These design choices can significantly improve the building's energy performance [16]. As window opening plays an important role in air circulation and sunlight acceptance its behavior also effects on energy consumption of the building. The impact of building insulation and window opening airflow control on the behavior of residents as it relates to HVAC energy consumption in Brazilian residential structures is assessed by the authors. Medium-sized buildings are more likely to offer their occupants thermal convenience when the proper building ventilation management is used. Additionally, proper building ventilation is achieved by automatic ventilation control combined with medium thermal inertia to lower HVAC system energy usage [34]. To preserve visual comfort while integrating manufactured and natural light, an estimation of energy consumption and visual efficiency was conducted for three distinct styles of glazed windows [35].

2.6 Studies on window orientation and energy efficiency

Window can also be a major factor in the energy consumption of the building. Several research studies were done to find the role of window orientation in energy consumption involves examining its influence on solar heat gain, daylighting, and energy use for heating and cooling, comparing different orientations (north, south, east, west) for their impact on energy efficiency, and considering seasonal variations and geographical factors for optimal orientation. The research examined how office buildings in various seasons use energy for lighting, heating, and cooling through the effects of room layout, window size, and orientation. The concept of an ideal window area underscores the potential for energy savings in rooms with larger windows, as they allow natural light to enter the space, thereby reducing the reliance on artificial lighting [36]. Research in cool climate regions like Lithuania has demonstrated that north-facing glazing can be an energy-efficient choice for air-conditioned office buildings without shading. Additionally, factors such as window-to-wall ratios, window orientation, and glazing type can significantly impact a building's overall energy consumption [37]. Using EnergyPlus and DAYSIM to generate charts displaying the most effective variable combinations, the effects of window size, orientation, and glazing properties on space heating demand, daylighting, and temperature surroundings were investigated. The results show that whereas low U-values are necessary for both north- and south-oriented rooms to decrease space warming needs, south-oriented windows should be thoughtfully constructed to prevent excessive heat,

with solar control coverings being a preferable choice to dynamic solar shadings, and that deep or narrow south-oriented rooms may find it difficult to attain adequate sunlight amounts without excessive heat [38]. According to research, it is less critical for the window area to face north, allowing architects greater flexibility in orienting the structure and determining window placement [39].

From the above literature review and several other case studies, it is evident that there is a lack of awareness about this type of research in village areas, and no case studies on energy consumption have been conducted in the rural regions of Sangli, Maharashtra. This article guides on selecting windows by evaluating the impact of window area and type on energy consumption.

3 Methodology

Data was initially collected from a range of sources, including AutoCAD 2D drawings, input from engineers and architects, and relevant existing documentation. A comprehensive 3D model of the building was then developed using Autodesk Revit, incorporating precise dimensions, materials, and construction details. Four unique scenarios were created to assess the impact of different building orientations (such as north, south, east, and west) on energy consumption. Furthermore, four distinct window scenarios were devised to explore the effects of varying window sizes, types, and orientations.

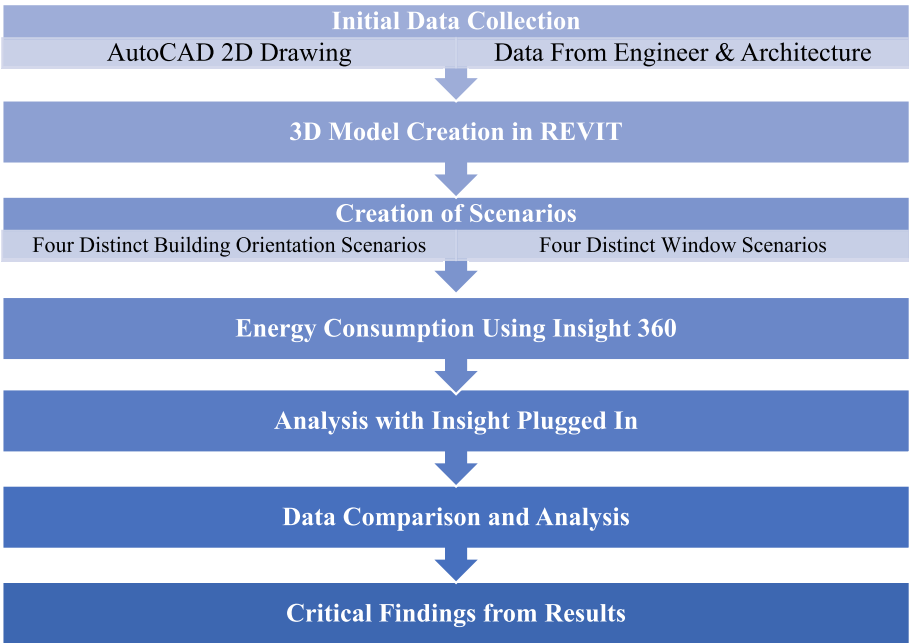
Energy consumption data was gathered using Insight 360, a robust energy simulation tool. The collected data was subsequently examined using Insight 360’s integrated analysis tools to identify trends and correlations.

3.1 General

The project was completed following the step-by-step procedure outlined in the flow chart above in Fig. 5. First, a three-dimensional (3D) model of the bungalow was created using Autodesk Revit Architecture software, which was based on AutoCAD 2D drawings provided by the bungalow’s engineer. This model served as the foundation for subsequent Building Energy Modeling (BEM) simulations. To comprehensively assess the impact of window conditions on various analyses, four distinct scenarios as shown in Table 1 were formulated, each with varying window configurations.

The simulations involved changing the position of windows as per given cases (Table 1) and rotating the building model at four cardinal orientations: 0°, 90°, 180°, and 270° relative to North. After the rotation was done, analysis was achieved using the Insight 360 plugin within Revit, which enabled the evaluation of energy consumption under different orientations and window setups. The BEM simulations were designed to calculate and compare energy use for each case, providing insights into how orientation and window conditions affect overall energy efficiency.

Fig. 5 Framework of research methodology



In addition to energy consumption, several analyses were conducted to gain a comprehensive understanding of the building's performance: Illuminance Analysis measured the distribution of light within the building to optimize daylight use and reduce reliance on artificial lighting; Solar Access Analysis evaluated the solar radiation received throughout the day and seasons to understand its impact on heating and cooling energy consumption; and Daylight Autonomy Analysis assessed how well daylight alone could meet the building's lighting needs. By comparing these results, potential improvements in building design and window configurations were identified to enhance energy efficiency, reduce overall consumption, and improve natural lighting conditions.

The following types of analysis need to be performed on the Revit model:

1. Illuminance analysis:

The amount of daylight which is obtained from the window maintains an internal environment makes visual comfort easy and helps to reduce the energy consumption of the building.[40]. These analyses are conducted to assess the building's illuminance levels, which depend on the size and type of windows and the amount of sunlight they allow in.

2. Daylight Autonomy: The percentage of the project surface with luminance above the design illumination is known as daylight autonomy [41].

3. Solar Access: The term solar access describes a building's capacity to receive sunlight in certain parts of the building, including its windows, roof, or solar panels.

4 Case study

4.1 Description

A G + 2 + basement residential bungalow (Fig. 6) located in a rural area of Sangli, Maharashtra, India, at latitude 16.985617 and longitude 74.129868, faces northeast and experiences a tropical hot-dry climate with temperatures ranging from 40 to 42 °C in summer to 14–16 °C in winter. The 650 m² structure, housing a family of five, features a long rectangular floor plan, lifts, reinforced concrete floors, Burnt Clay Brick walls, a bitumen access road, basement parking, and coverings of plaster, painted plaster, and ceramic cladding in the toilets. A detailed study of climate conditions is necessary while we plan to achieve energy efficiency. Western Maharashtra's climate is significantly influenced by its geographical location and seasonal monsoon patterns. The region experiences a tropical climate with distinct wet and dry seasons. Summers, from March to May, are characterized by hot and dry conditions, with temperatures often exceeding 40 °C in the interior plains. The monsoon season, spanning June to September, brings substantial rainfall, primarily from the southwest monsoon. The Western Ghats act as a barrier, causing orographic rainfall in the hilly areas, making them significantly wetter than the plains to the east. Post-monsoon, the winter season from October to February offers mild and pleasant weather, with temperatures ranging between 10 and 30 °C. The tabular form of the details of proper is given below (Table 2).

Table 1 Examined cases under the model

Case No	Type of window fixed in the Model	Size of window (mm)	Type of glass material	No. of windows
1	Original model	Original model	Original model	Original model
2	Transom	1524 × 3048	Low-E, e = 0.1	3 each on East& West only
3	Casement double	850 × 900	Low-E, e = 0.1	6 each on East& West only
4	No changes	No changes	Low-E, e = 0.1	No changes

Fig. 6 3D model of the bungalow



Table 2 Details of case study

Property	Details
Building type	Residential bungalow
Location	Sangli, Maharashtra, India
Story height	3.0 m
Floor area	650 m ²
Building height	9 m
Latitude	16.985617
Longitude	74.129868
GPS coordinates	16° 59' 8.2212" N, 74° 7' 47.5248" E
Temperature range	14–16 °C (winter) to 40–42 °C (summer)
Climate	Tropical hot-dry climate



Fig. 7 Different scenarios of windows for cases 1, 2, 3 & 4 respectively

5 Data analysis

The results were checked for each of the four cases (Fig. 7) by rotating the building orientation from North to 90°, 180°, 270°, and 360° angles, considering variations in window type, size, number, glass material, and orientation, and then analyzing the corresponding energy consumption to observe the impact of improvements in these scenarios. The window placement adjustments based on the specific case include:

Table 3 Key findings resulting from the analysis (Case 1)

Orientation	Illuminance analysis (%)			Daylight autonomy (%)	Solar access (%)	Energy consumption (kWh/m ² /yr)
0°	07:00 HRS	73	53	3	70	333
	09:00 HRS	54				
	11:00 HRS	56				
	13:00 HRS	47				
	15:00 HRS	46				
	17:00 HRS	41				
90°	07:00 HRS	46	50	3	80	331
	09:00 HRS	39				
	11:00 HRS	56				
	13:00 HRS	45				
	15:00 HRS	61				
	17:00 HRS	51				
180°	07:00 HRS	52	47	3	75	337
	09:00 HRS	44				
	11:00 HRS	48				
	13:00 HRS	48				
	15:00 HRS	55				
	17:00 HRS	36				
270°	07:00 HRS	52	51	5	85	320
	09:00 HRS	47				
	11:00 HRS	51				
	13:00 HRS	53				
	15:00 HRS	63				
	17:00 HRS	38				

5.1 Case 1

The original position of the windows with a normal type of glass according to the real data obtained. In this case, the original 3D model has been created without changes in the size, type, and number of windows. Here are the findings for various orientations (Table 3):

Based on observations and calculations across different orientations, the optimal orientation is determined to be 90°, ensuring proper daylight autonomy, solar access at 270°, low energy consumption, and optimal illuminance analysis results at 0°.

5.2 Case 2

Three existing windows were replaced with three casement windows, each positioned on the East and West sides of the bungalow, which faces North, featuring double 1524 mm x 3048 mm dimensions and low-emissivity glass ($e=0.1$). Here are the findings for various orientations (Table 4):

Based on observations and calculations of various orientations, the 90° orientation proves optimal, offering proper daylight autonomy at 270°, minimal solar access at 0°, low energy consumption at 180°, and optimal results for illuminance analysis at 90°.

Table 4 Key findings resulting from the analysis (case 2)

Orientation	Illuminance analysis (%)			Daylight autonomy (%)	Solar access (%)	Energy consumption (kWh/m ² /yr)
0°	07:00 HRS	73	53	3	70	291
	09:00 HRS	57				
	11:00 HRS	56				
	13:00 HRS	47				
	15:00 HRS	45				
	17:00 HRS	41				
90°	07:00 HRS	62	49	32	80	284
	09:00 HRS	50				
	11:00 HRS	55				
	13:00 HRS	45				
	15:00 HRS	45				
	17:00 HRS	39				
180°	07:00 HRS	53	48	2	75	257
	09:00 HRS	45				
	11:00 HRS	50				
	13:00 HRS	50				
	15:00 HRS	55				
	17:00 HRS	35				
270°	07:00 HRS	52	50	6	80	322
	09:00 HRS	46				
	11:00 HRS	49				
	13:00 HRS	50				
	15:00 HRS	63				
	17:00 HRS	38				

5.3 Case 3

The three existing windows on each of the East and West sides were replaced with a total of twelve smaller windows measuring 850 mm x 900 mm, featuring low-emissivity glass with $e = 0.1$, and one existing window was replaced with two windows, resulting in six windows on each side of the bungalow. Here are the findings for various orientations (Table 5):

Based on observations and calculations, a 90° orientation is ideal for daylight autonomy, 270° for solar access with low energy consumption, and 0° for the best results in illuminance analysis, making 90° the best overall orientation for the case.

5.4 Case 4

We have substituted all current windows, originally fitted with standard glass, with low-emissivity glass rated at $e = 0.1$. Here are the findings for various orientations (Table 6):

Based on observations and calculations, 0° orientation is ideal for daylight autonomy, 90° and 270° orientations are optimal for solar access, and 270° orientation also shows low energy consumption, making 90° the best overall orientation for the case.

6 Comparative analysis

6.1 Illuminance analysis

The following common settings were consistently maintained in the Revit Insight plugin for all time settings to obtained the results as shown in Fig. 8.

Table 5 Key findings resulting from the analysis (case 3)

Orientation	Illuminance analysis (%)			Daylight autonomy (%)	Solar access (%)	Energy consumption (kWh/m ² /yr)
0°	07:00 HRS	71	52	35	70	335
	09:00 HRS	55				
	11:00 HRS	57				
	13:00 HRS	47				
	15:00 HRS	45				
	17:00 HRS	39				
90°	07:00 HRS	61	50	34	80	331
	09:00 HRS	49				
	11:00 HRS	56				
	13:00 HRS	47				
	15:00 HRS	45				
	17:00 HRS	39				
180°	07:00 HRS	51	49	2	75	339
	09:00 HRS	45				
	11:00 HRS	52				
	13:00 HRS	50				
	15:00 HRS	56				
	17:00 HRS	37				
270°	07:00 HRS	51	49	5	85	319
	09:00 HRS	48				
	11:00 HRS	50				
	13:00 HRS	49				
	15:00 HRS	58				
	17:00 HRS	38				

- Sky model: Perez All-Weather Sky;
- Threshold: Lower and upper: 300 and 3000 lx;
- Analysis plane Height: 12 inches above the floor;
- Clear sky.

The results of the Illuminance Analysis for a 0° orientation are displayed in Fig. 9. By observing the above data in case of illuminance analysis 0° orientation for all 4 cases gives the best results.

6.2 Daylight autonomy

Figure 10 shows Percentage of Daylight Autonomy for Different Orientations & Cases in Analysis.

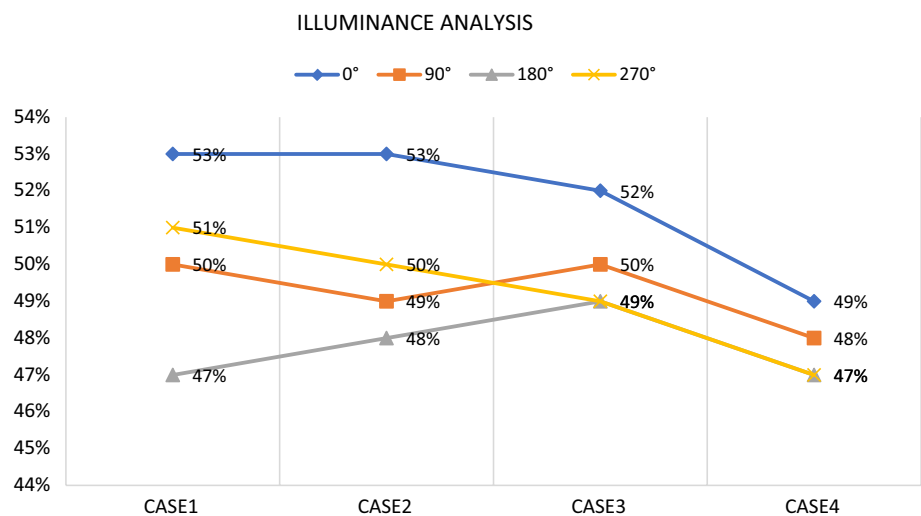
sDA: sDA300/50 (> 300 lx, > 50% of year). The percentage of floor area that receives over 300 lx for at least 50% of 3650 annual hours.

ASE: ASE1000/250 (> 1000 lx, < 250/3650 of the year). The percentage of floor area that receives over 1000 lx for more than 250/3650 annual hours.

The results of the Daylight Autonomy for a 0° orientation are displayed in Fig. 11. Building scored 0 LEED points for all 4 cases. From the above readings, 0° orientation of both cases 3 & 4 have obtained 0 LEED points with 35% of the building area meeting sDA % hours in rooms with less than 20% area above ASE. At least 55% of the rooms must be larger than sDA300/50 if ASE1000/250 makes up less than 20% of the total room area. However, the following results do not meet the satisfactory criteria.

Table 6 Key findings resulting from the analysis (case 4)

Orientation	Illuminance analysis (%)			Daylight autonomy (%)	Solar access (%)	Energy consumption (kWh/m ² /yr)
0°	07:00 HRS	65	49	35	70	333
	09:00 HRS	53				
	11:00 HRS	52				
	13:00 HRS	43				
	15:00 HRS	43				
	17:00 HRS	36				
90°	07:00 HRS	59	48	34	80	331
	09:00 HRS	50				
	11:00 HRS	53				
	13:00 HRS	42				
	15:00 HRS	43				
	17:00 HRS	38				
180°	07:00 HRS	51	47	34	75	337
	09:00 HRS	45				
	11:00 HRS	50				
	13:00 HRS	48				
	15:00 HRS	53				
	17:00 HRS	34				
270°	07:00 HRS	52	47	4	80	320
	09:00 HRS	47				
	11:00 HRS	48				
	13:00 HRS	44				
	15:00 HRS	53				
	17:00 HRS	36				

Fig. 8 Graph with percentage of illuminance for different orientations & cases in illuminance analysis

6.3 Solar access

Determining whether an area gets adequate natural light during the entire day is possible with the help of solar access analysis. It is also known as daylight access, and solar access standards like Australia's SEPP 65 are becoming mandatory for new construction in many nations and areas [42]. This kind of investigation determines how many

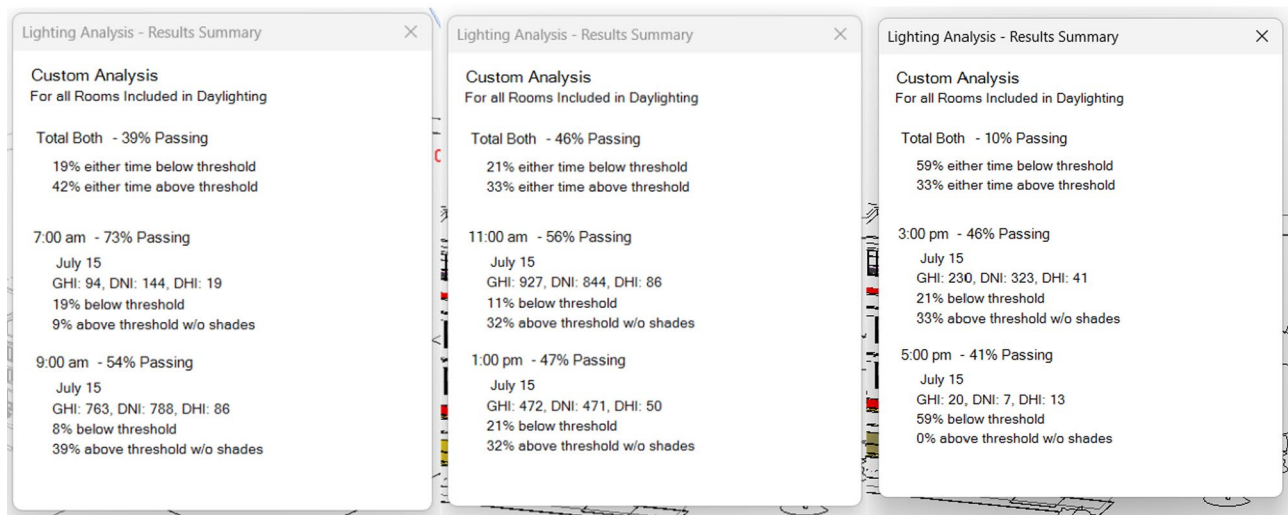
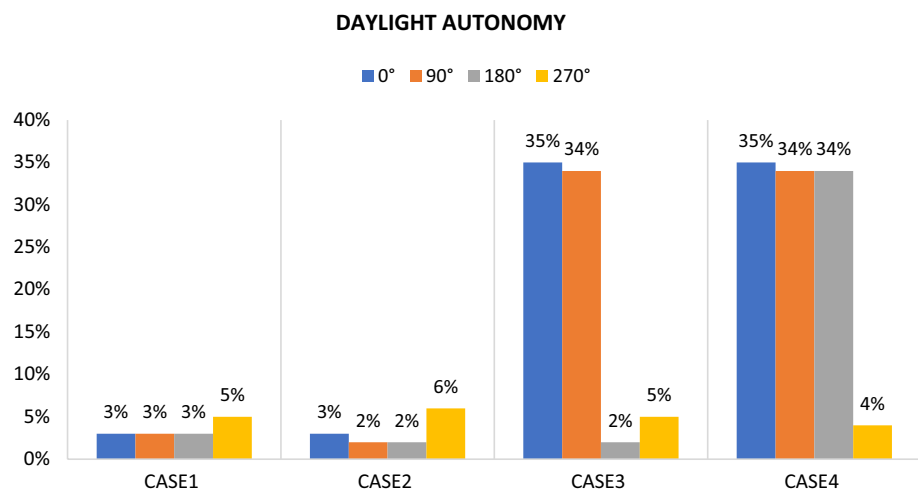


Fig. 9 Outcomes for illuminance analysis at a specific time for case 1's 0° orientation

Fig. 10 Percentage of daylight autonomy for different orientations & cases in analysis



hours of a particular day, as well as a specific range of hours within that day, that a space is affected by direct sun radiation. The Solar Access data are presented on a graph in Fig. 12.

- Date/Time: 07/24 from 7:00 HRS to 17:00 HRS (10 h).
- Threshold: minimum of 3 h.

From 7:00 HRS to 17:00 HRS—85% of Rooms Analyzed meet a minimum number of hours of Direct Solar Passing.
 85% of rooms have ≥ 3 h.
 15% of rooms have 0 h.

Observing the above data in the case of solar access 270° orientation for all 4 cases gives the best results. The model view of Solar Access Analysis is shown in Fig. 13.

6.4 Energy consumption

The data indicates that a 270° orientation yields better energy consumption for all four cases as shown in tabular format in Fig. 14. Figure 15 shows the outcomes of Insight 360's consumption of energy for Case 1.

From the above analysis, the optimal orientation for illuminance is 0° for cases 1 and 2, with a large transom window with low-E glass being an effective solution; daylight autonomy improves with low-E glass in cases 3 and 4, with the best

Fig. 11 The outcome obtained for 0° in daylight autonomy analysis case 1

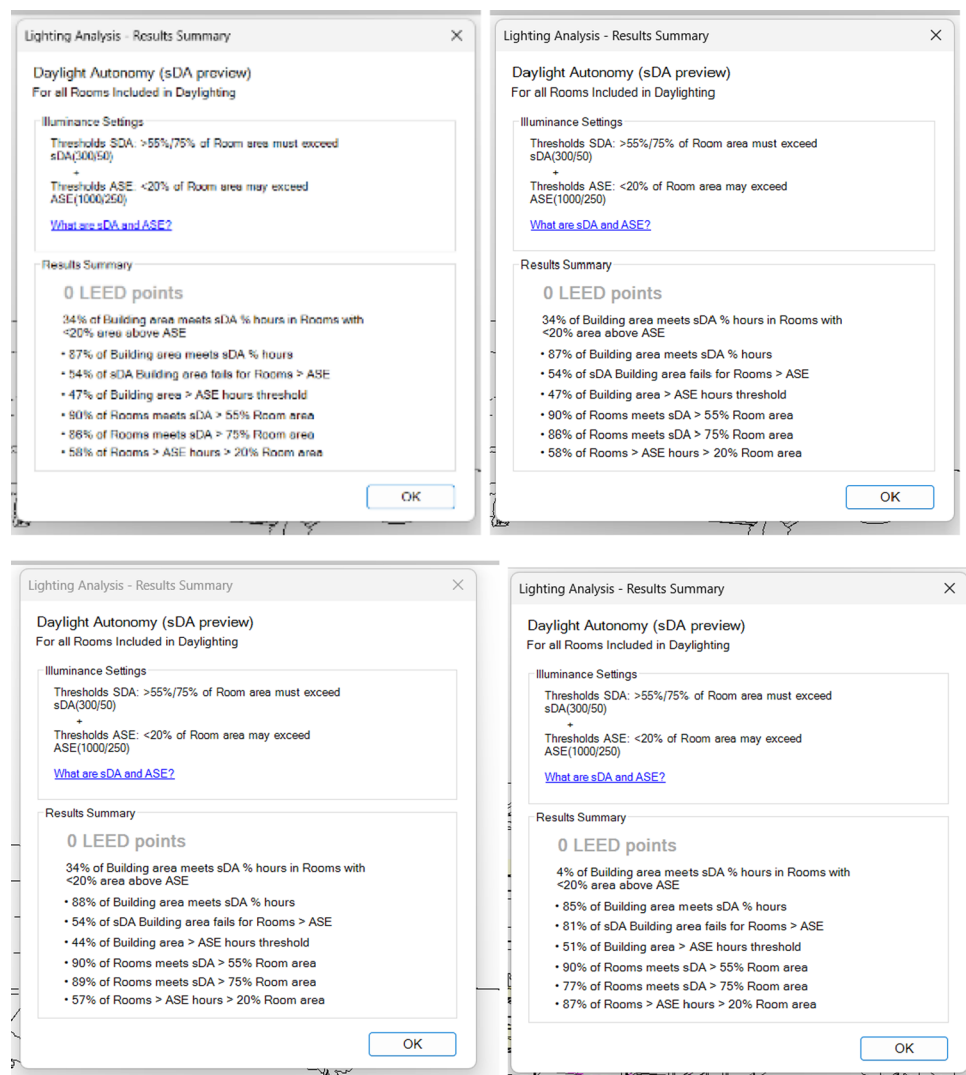
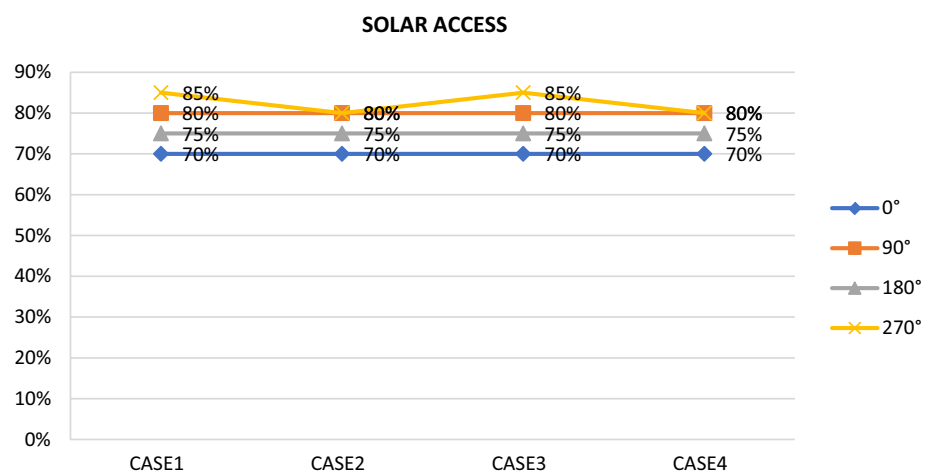


Fig. 12 Percentage of solar access for different orientations & cases in analysis



orientations being 0° and 90° for small windows; for solar access, 0° is best in cases 1 and 3, and 90° works well in all cases; and for energy consumption, the best orientations are 90° and 180° in case 2, and 90° and 270° in case 3. The analysis results of Insight 360's consumption of energy for Case 1 are shown in Fig. 14. In India, the location of village is such that

Fig. 13 Revit insight model view of solar access analysis

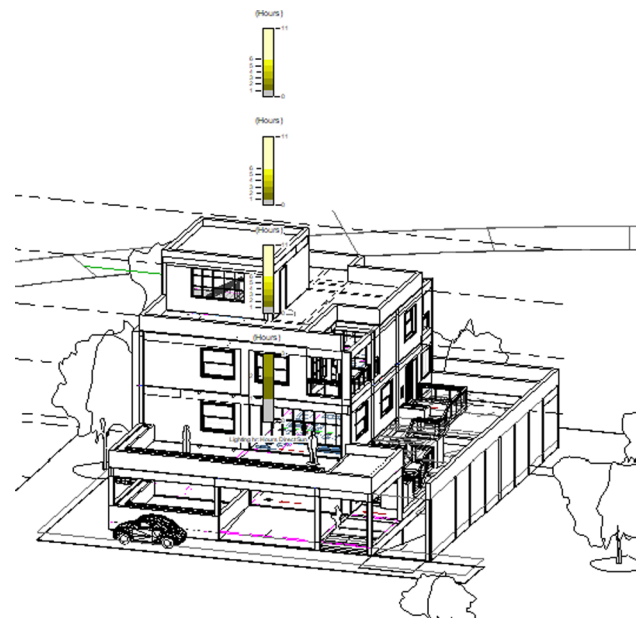
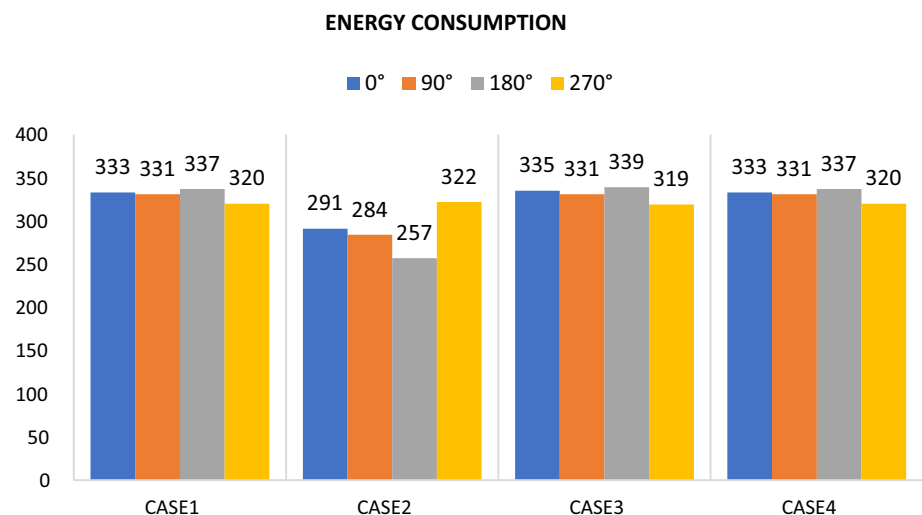


Fig. 14 Comparison of values for different orientations & cases in analysis



though sunrise and sunset happen in East and West directions respectively, there is a minor tilt of the sun's path towards the northern due to the tilt of the earth. Usually, the weather is typically hot and dry, with high temperatures. So, here all buildings must keep their facing of the major axis of the building towards the North to avoid excessive sunlight and heat to avoid heating and cooling load.

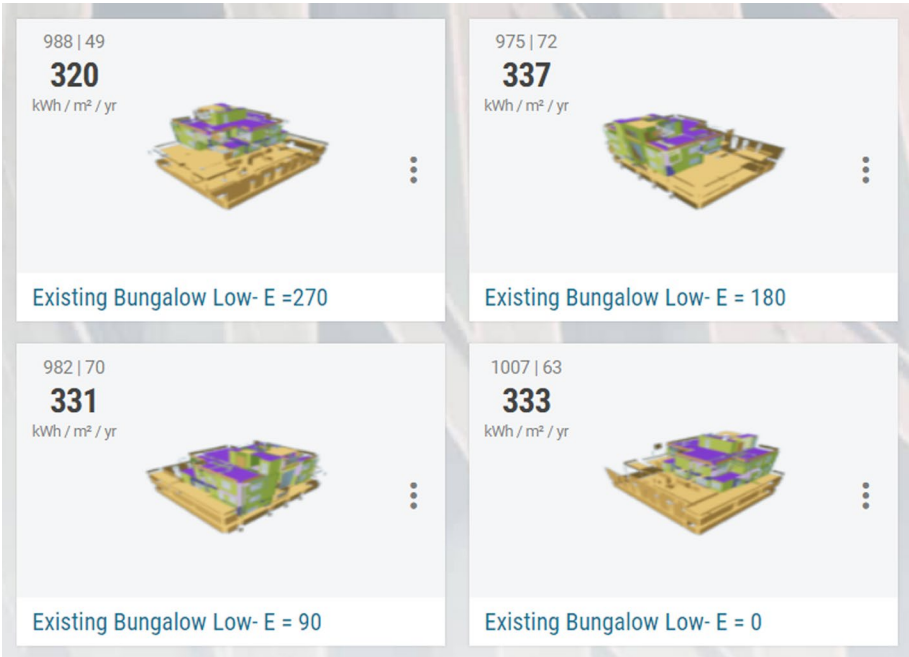
In the model Room 3 has been considered to study the impact of heating and cooling and the description of it is given as follows:

Zone Load Summary for Cooling of Room 3 in Model.

Figure 16 shows the Zone Load Summary for Cooling in Room 3 of model. This analysis probes into the cooling requirements for a specific three-room zone within a building. Maintaining a comfortable temperature of 23.9 °C necessitates a total cooling load of 2,529 watts. Interestingly, heating demands are practically negligible in this scenario. Examining the cooling load breakdown reveals that solar heat gain through windows is the primary culprit, contributing a hefty 55.3% (1,399 watts). Conduction gains (182 watts) and infiltration (338 watts) follow closely behind, collectively accounting for 76.6% of the total load.

Internal heat sources also play a role, with occupants generating 130 watts, while lights and equipment add 97 watts and 127 watts respectively, bringing their combined contribution to 14% of the overall load. Ventilation systems chip in with an additional 9.5%, primarily driven by the zone's need for fresh air (240 watts).

Fig. 15 Outcomes of insight 360's consumption of energy for case 1



Zone Load Summary

3 Room COOLING

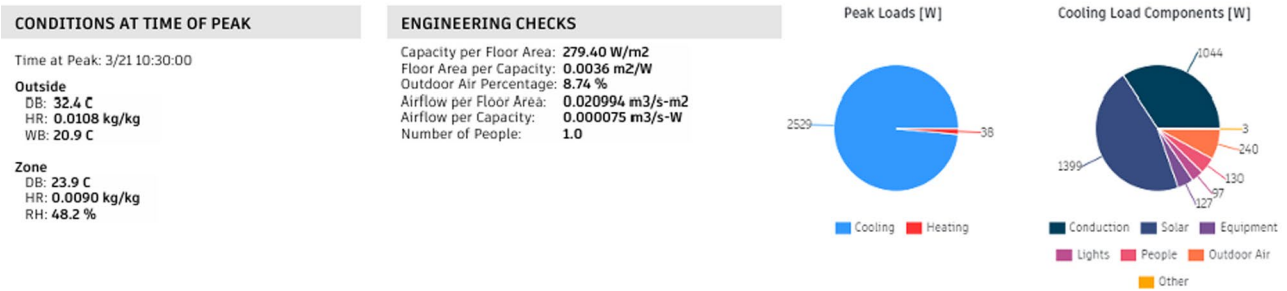


Fig. 16 Zone load summary for cooling in room 3 of model

Zone Load Summary

3 Room HEATING

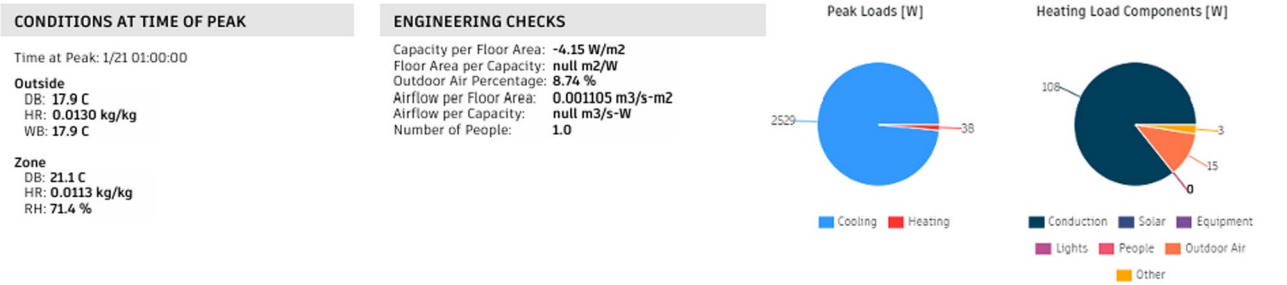


Fig. 17 Zone load summary for heating in room 3 of model

The analysis emphasizes the significant role the building envelope plays in energy consumption due to solar radiation and air infiltration. Strategic window design to minimize solar heat gain and improved insulation and airtightness are highlighted as potential areas for significant reduction in cooling energy demands. Additionally, effective ventilation management, particularly in maintaining adequate fresh air flow, contributes to overall energy efficiency.

In conclusion, this zone experiences a peak in cooling demand due to a combination of factors: solar heat gain, infiltration, and internal heat generated by equipment and occupants. The total cooling requirement of 2,529 watts sheds light on the key contributors to energy consumption and pinpoints areas where efficiency improvements can be made. This information is crucial for determining the appropriate size for the HVAC system, optimizing its design, and formulating effective building energy conservation strategies.

Zone load summary for Heating of Room 3 in model.

Figure 17 illustrates the Zone Load Summary for heating in Room 3 of the model. The heating load analysis for the "Room 3" zone reveals a remarkable energy efficiency. During the peak heating period on January 1st, the zone experienced a net heat surplus rather than requiring additional heating. This surplus is primarily attributed to the building's envelope and ventilation system. Infiltration of warmer outside air and efficient heat conduction through the walls and ceiling significantly reduced the overall heating demand. Furthermore, the ventilation system played a crucial role in balancing heat losses with latent heat gains, contributing to a comfortable internal environment without the need for supplemental heating. These findings demonstrate the effectiveness of the building's design in optimizing energy consumption and providing a comfortable living space.

7 Impact of paint on energy consumption

While factors like insulation and building materials are paramount, the impact of paint colour on house temperature is often underestimated. Understanding this relationship can enable informed decisions about heat-reflective paint colors, contributing to a more comfortable and energy-efficient living space during hot weather. Further study is essential to gain more insight on the impact of colours and other parameters.

8 Limitations to achieving energy-efficient buildings in terms of socioeconomic factors

- The coastal areas, like those near Mumbai and Konkan, are humid year-round due to their proximity to the Arabian Sea, while the inland areas, such as Pune and Kolhapur, have a more moderate climate with lesser humidity. The material used in the building should satisfy and meet the requirements so as this sustains from all climatic conditions and provide better comfort and ease the standard of living. This further extends to the appropriate selection of locally available materials and cost-effective architecture
- The impact of social problems on green buildings, such as the basic understanding of customers, purchase intention, social acceptance, social needs, public attitudes, and behaviours affect the consumers' decision-making process. These need to be considered while planning for energy-efficient buildings as these may hamper the decision of consumers to select energy-efficient designs and models.
- Factors such as space layout, thermal conditions, noise levels, lighting, and air quality in indoor environments must be measured.
- Feasibility studies need to be carried out to understand the availability of locally available construction materials and an they are cost-effective and sustainable as well as energy efficient to sustain all weather conditions and for the long term by keeping a futuristic approach.
- Agriculture is a significant part of the lifestyle, with sugarcane, grapes, and pomegranates being the primary crops. Also the lifestyle of people needs to be considered as the majority of the population resides in rural areas and approximately 33% in urban regions of Sangli district. This shall help us to understand whether people are oriented towards energy-efficient residential buildings, consumer requirements, and needs. Also, it will give us an overall background of the average economic condition of the area which shall further make way to develop or design strategies to overcome these problems by effectively using passive strategies and cost-effective techniques.
- Further data collection is necessary in terms of the life of people and their socio-economic status while planning for energy-efficient buildings. This will help us to understand and plan the buildings and construction process in an economical way

9 Conclusion

The reading concludes that a building oriented at 0° north provides optimal analysis results for the region, with Low-E glass and smaller windows preferred for similar climates, suggesting east and west window orientations and placing key rooms like the living room facing north, while for energy consumption, a 90° orientation and large windows are best, thus recommending an overall 0°–90° orientation and case 3 among four cases, with future research to explore additional design parameters like geometry and roof shapes for further energy optimization.. A limitation of this study is the exclusion of the impact of adjacent structures on the building's energy performance. Future research should consider the shading effects and airflow patterns created by neighboring buildings, as these factors can significantly influence energy consumption and overall building efficiency. In conclusion, enhancing a building's energy performance necessitates a comprehensive approach that considers various design factors, including thermal comfort, lighting, ventilation, and the occupants' functional and aesthetic needs. Future research is essential to develop innovative optimization methods and tools that can address these challenges while empowering architects to design energy-efficient structures. Additionally, expanding the optimization process to include additional parameters, such as building orientation, shading devices, and multi-story configurations, can further contribute to the creation of sustainable and efficient buildings.

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Data availability The data that support the findings of this study are available from the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

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